

RMO(INDIA) – 1993

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABC be an acute angled triangle and CD be the altitude through c. If $AB = 8$, $CD = 6$, find the distance between the mid points of AD and BC.
 2. Prove that ten's digit of any power of 3 is even.
 3. Suppose $A_1A_2\ldots A_{20}$ is a 20 sided regular polygon. How many non-isosceles (scalene) triangle can be formed whose vertices are among the vertices of the polygon but whose sides are not the sides of the polygon?
 4. Let ABCD be a rectangle with $AB = a$, $BC = b$. Suppose r_1 is the radius of the circle passing through A and B and touching CD; and similarly r_2 is the radius of circle passing through B and C and touching AD. Show that $r_1 + r_2 \geq \frac{5}{8}(a+b)$.
 5. Show that $19^{93} - 13^{99}$ is a positive integer divisible by 162.
 6. If a, b, c, d are four positive real numbers such that $abcd = 1$, prove that $(1+a)(1+b)(1+c)(1+d) \geq 16$.
 7. In the group of ten persons, each person is asked to write the sum of the ages of all the other nine persons. If all the ten sums form nine element $\{82, 83, 84, 85, 87, 89, 90, 91, 92\}$, find the individual ages of the persons, assuming them to be whole numbers (of years).
 8. I have six friends and during certain vacation I met during several dinners. I found that I dined with all six exactly on one day; with every five of them on 2 days; with every four of them on 3 days; with every three of them on 4 days; with every two of them on 5 days. Further every friend was present at 7 dinners and every friend was absent at 7 dinners. How many dinners did I have alone?

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